Final Project Part 1

- Although the final project is a group project, but for part 1, **every** team member needs to complete the following tasks:
  - **Install** development environment (Jupyter Notebook or Google Colab + Pytorch) – slides 2-4
  - **Download** MNIST handwritten digits image dataset – slides 7-8
  - **Prepare** your own set of handwritten digits image collections following specified file name convention – slides 5-6
  - **Upload** your images to a public Github repository – slide 6

Final Project Part 1 – Task Overview

- **Install Pytorch**
  - Regular laptop is good enough for the training of the neural network model we will build.
  - You are welcome to use more powerful computing platform you have access to.
- **Download MNIST Dataset**
  - Study the image data in the dataset
  - Figure out how can create your own dataset of same kind using handwritten digit by your group members.

- **Everyone creates your own handwritten digit dataset. Follow the same format as MNIST Dataset**
  - 28x28 Grayscale
  - White handwritten digit image on black background (may need some image processing)
  - At least one image for each digit
  - At least one set from each team member
- **Submit your group's handwritten digit image set through github (see page 6)**

Jupyter Notebook or Google Colab

You can use either Jupyter Notebook

- Local installation required → install Anaconda, you will have Jupyter Notebook (follow link on the right side)

Or Google Colab

- Cloud (browser based)
- Local installation ⇒ follow link on the right side

Good installation guide for Anaconda → JupyterNotebook → Google Colab
https://saturncloud.io/blog/how-to-run-google-colab-locally-a-step-by-step-guide/

Colab notes:

- Browser-based Colab session stops when you go idle.
- Colab Pro may run in the background.

Pytorch Installation

- MacOS, Terminal command line:
  - `conda install pytorch torchvision -c pytorch`
  - `python -c "import torch; print(torch.__version__)"`
- Windows: https://saturncloud.io/blog/how-to-install-pytorch-on-windows-using-conda/

Official Installation Instruction:
https://pytorch.org/get-started/locally/

May need matching of versions:
https://pypi.org/project/torchvision/

torch	torchvision	Python
main / nightly	main / nightly	>=3.8, <=3.11
2.1	0.16	>=3.8, <=3.11
2.0	0.15	>=3.8, <=3.11
1.13	0.14	>=3.7.2, <=3.10

Create Test Images

1. Complete collection of handwritten digit data set.
    - One image for each of 0-9 digits handwritten by each team member.
    - Black background, white handwritten digits
    - 28x28, gray scale image
2. Collected dataset should be uploaded to the specified Github repository with specified filename format (more detail in the next page)
3. Submit the following in Canvas:
    - A pdf or text file specifying **your groupID and memberID (**within your group – 1, 2, 3 etc)
    - Make sure you coordinate within your group to assign one ID per member.

Optional:

- Investigate the format of the MNIST dataset downloaded from the reference Jupyter notebook code (in Canvas/Modules/JupyterNotebooks)

Image and File Name Format

Format File Names

- Use png files to store the handwritten digit images
- Filename format for team collected digit images:

  <label/digit>-<groupID>-<memberID>.png
- For example, image of digit 3 by group 4 member 1, it is 3-4-1.png

Upload to this public Github repository:
https://github.com/boshen-csuchico/Handwritten-Digits-Fall-2026.git

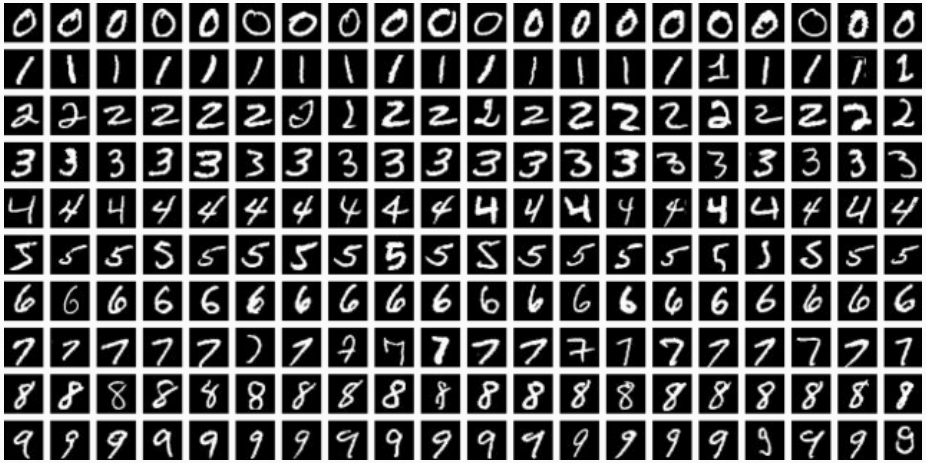
- Fork from the repository (use Pull Request)
- Insert your image files in the /digits/ folder
- Create Push Request (PR) from your fork
- **Either every group member does the above or have one member to upload for all.**
- Wait for instructor's review and merge

Optional:
Experiment with Loading from MNIST

Sample Pytorch code:

```
trainset =
datasets.MNIST('~/MNIST_data/',
download=True, train=True)

trainloader =
torch.utils.data.DataLoader(trainset,
batch_size=64, shuffle=True)
```



PyTorch Dataloader

10000*28*28=7840000

```
7840016 May 23  2020 t10k-images-idx3-ubyte
  10008 May 23  2020 t10k-labels-idx1-ubyte
47040016 May 23  2020 train-images-idx3-ubyte
  60008 May 23  2020 train-labels-idx1-ubyte
```

MNIST

60000*28*28=47040000

CLASS torchvision.datasets.MNIST(*root: str, train: bool = True, transform: Optional[Callable] = None, target_transform: Optional[Callable] = None, download: bool = False*) [SOURCE]

MNIST Dataset.

Parameters:

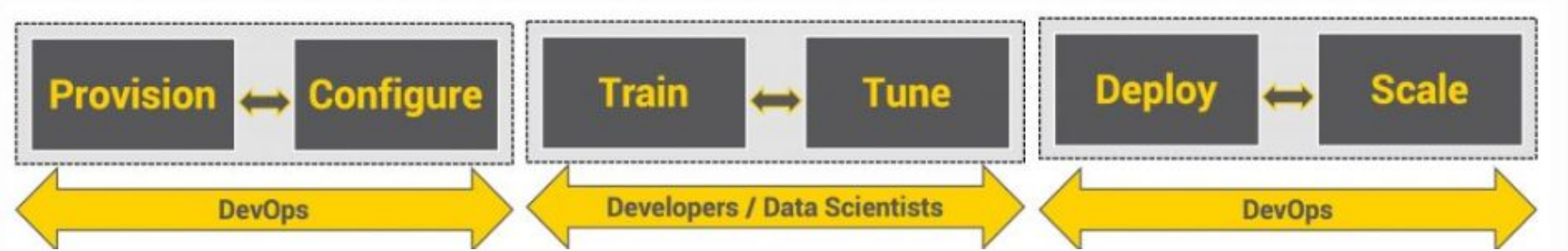
- **root** (*string*) – Root directory of dataset where `MNIST/raw/train-images-idx3-ubyte` and `MNIST/raw/t10k-images-idx3-ubyte` exist.
- **train** (*bool, optional*) – If True, creates dataset from `train-images-idx3-ubyte`, otherwise from `t10k-images-idx3-ubyte`.
- **download** (*bool, optional*) – If True, downloads the dataset from the internet and puts it in root directory. If dataset is already downloaded, it is not downloaded again.
- **transform** (*callable, optional*) – A function/transform that takes in an PIL image and returns a transformed version. E.g, `transforms.RandomCrop`
- **target_transform** (*callable, optional*) – A function/transform that takes in the target and transforms it.

Test dataset

Additional Information about Pytorch

ML Workflow
Conceptual vs. Production





ML Stack

- Google based
  - Language: Python + Numpy, (R is also supported)
  - Framework: Keras (Library)
  - Platform: Tensorflow
  - Infra: GCP

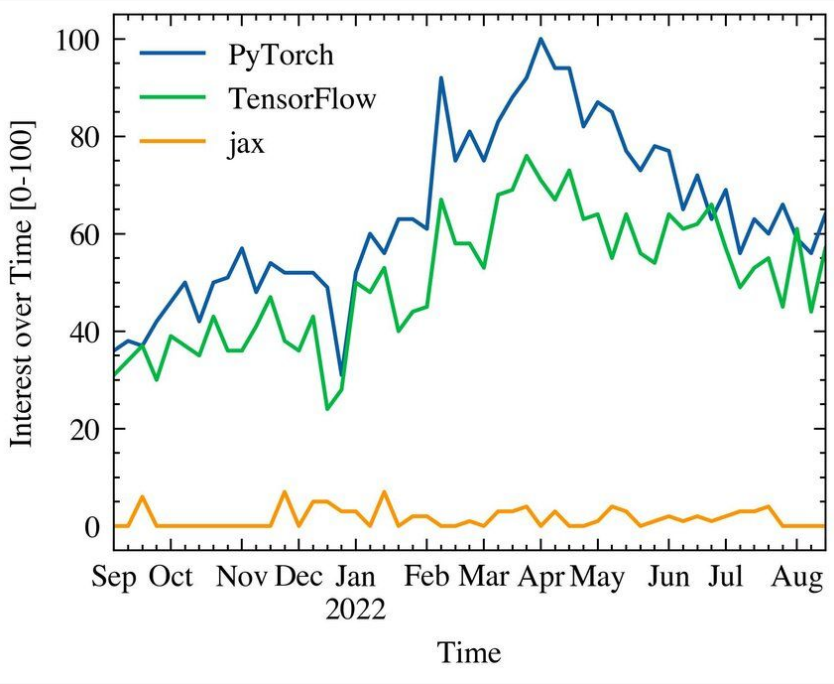
- Facebook based
  - Language: Python
  - Framework: Caffe2 (production), or Pytorch (research→ production)
  - Platform: FBLearner, ONNX (API)
  - Infra: Open Compute

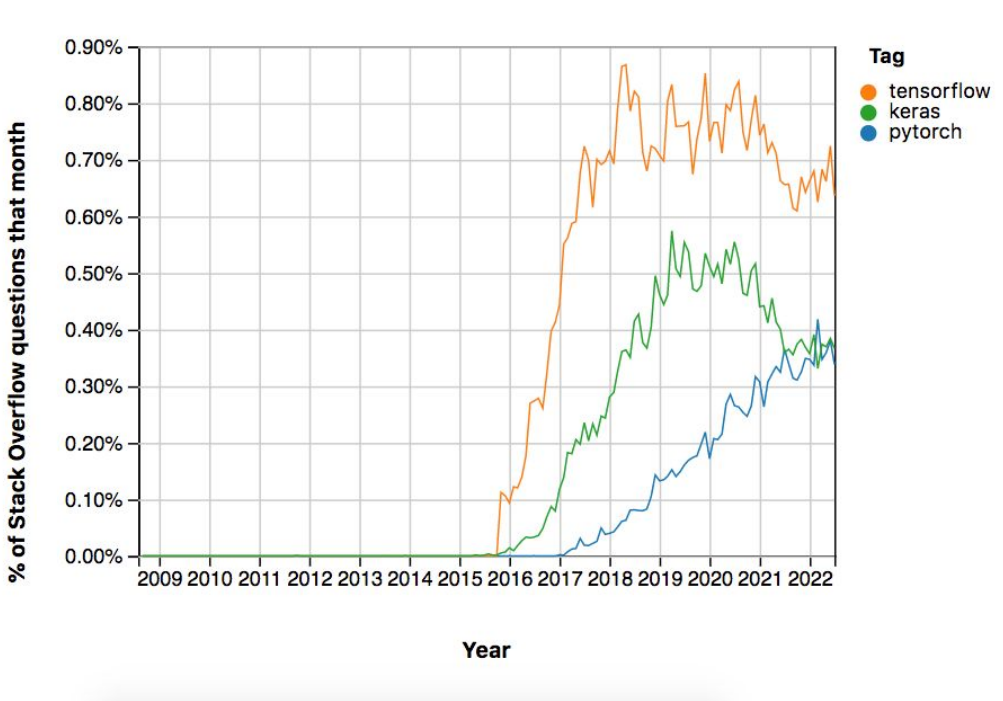
ML Stack 2

- Apache Stark based
  - Language: Scala, Java, SQL, Python, R
  - Framework: Databrick, (?MxNet?)
  - (Data flow) Platform: MapReduce/Hadoop → Spark (distributed computing library, include MLlib)
  - Infra: AWS, ...

- Microsoft based
  - Language: C#, F#, Python
  - Framework: CNTK → ML.NET
  - Platform: ONNX (API)
  - Infra: Azure

Popularity


PyTorch
TensorFlow
jax
Interest over Time [0-100]
100
80
60
40
20
0
Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug
2022
Time


% of Stack Overflow questions that month
0.90%
0.80%
0.70%
0.60%
0.50%
0.40%
0.30%
0.20%
0.10%
0.00%
2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022
Year
Tag
tensorflow
keras
pytorch

PyTorch in Research Community

